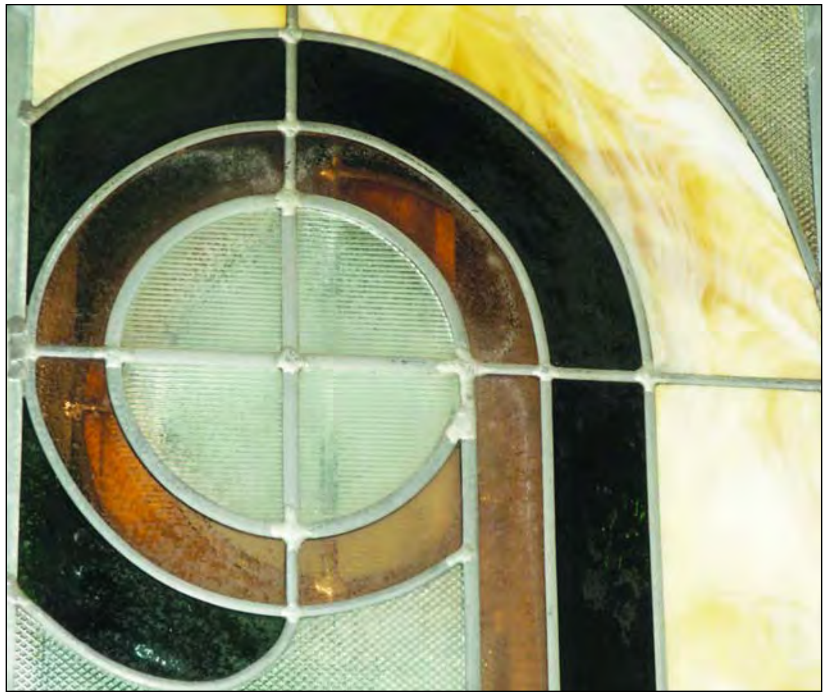
Taking It All Apart

Why demolish when used building materials can get recycled or resold?

by Miriam Lara Boon



LULU JONES

Traditional construction and demolition projects add to the landfills that dot our landscapes and, a drop at a time, oceans of waste are being created. An estimated 51.6 million tons of waste were generated by residential demolition and renovations in 1996, according to the EPA, and all construction waste amounted to 136 million tons. Peaks to Prairies Pollution Prevention Information Center estimated that this waste comprised 10%-30% of the total amount of refuse generated in the United States that year. Over the last decade or so, as awareness of this blight has grown, so have companies and programs designed to rethink our throw-it-all-away economy. Though some have found unique solutions to the question of how we can continue to build and renovate without contributing to the growing landfill problem, many are arriving at the same answer, at least for renovation projects—deconstruction.

"Deconstruction is the systematic disassembly of usually light-framed structures in the strict reverse order of construction," says Jim Primdahl of the Deconstruction Management Group, in Portland, Oregon. He couches the issue in terms of thermodynamics, discussing the need to extend the embodied energy of building materials. Deconstruction of your average single-family house can save up to 23 million Btu of embodied energy—equivalent to 205 gallons of gasoline—according to the Deconstruction Institute, a Web-based clearinghouse of deconstruction information, which is supported by the Florida Department of Environmental Protection.

By that token, stores that sell used building materials have been preserving embodied energy for years. These stores, operating on a nonprofit basis, acquire used building materials by several methods. "We have homeowners that are doing renovations or cleaning out garages, retail stores that are getting rid of old product, and small contractors that have extra product or used product that they bring to us as well," says Pat McLean of the Habitat for Humanity Canada's ReStores in Canada. ReStores also acquire used building materials by sending their own volunteer salvage team to construction sites at the request of site owners. Materials from these salvage missions make up about 20% of their stock. "It has been increasing slowly," adds McClean. "We don't take every project we're offered, either. It has to have enough material and be volunteer friendly—a safe kind of working environment."

The Sound Builder's ReSource in Olympia, Washington, runs a materials drop-off site at their local landfill from which they acquire some of their materials. Companies that donate materials to this and similar nonprofit stores are rewarded with tax write-offs. Many stores have a pickup service that will go to a site or home to get the material that is being donated or sold.

In the last three years, some of these operations have assembled teams to deconstruct entire houses in place of demolition. The results have been a pleasant surprise for everyone involved. "A typical 1,400 ft^2 Craftsman costs about $8,000 in labor to take down," explains Primdahl. The tipping fees for the materials that can't be reused might amount to $1,400—quite affordable when contrasted with the $4,000 in tipping fees that demolition would incur. In Portland, with this type of house, about 15% of the materials usually get sent to the landfill, 25% get recycled or otherwise diverted, and 60% get salvaged and resold for $7,000-$12,000.

With deconstruction, the percentage of materials diverted from the landfill but not resold—25% in the Portland case—is strongly influenced not only by the house's construction, but also by the recycling and processing facilities available in the region. Not all regions have the facilities needed. Materials that end up in the landfill include plaster; drywall; sometimes roofing material depending on available recycling facilities; and small and mixed debris that cannot be reused or recycled for a variety of reasons. Salvageable materials consist primarily of soft-skin materials—cabinets, doors, windows, flooring, fixtures, furnaces and other appliances, siding, and bricks. Within the skin, lumber is also salvageable. Stucco is the big disappointment. "It's hard to get off, it's labor intensive, and it's not even reusable," says Primdahl. It cannot even be recycled, because the quantities are too small, because other materials, such as lead paint or wire mesh, are usually mixed in with it, and because it usually must be removed from the site before the bulk of the concrete—the foundations—are torn up and sent to the recycling plant.

RESOURCE MANAGEMENT GROUP, INC.

By systematically disassembling a house, many of its components can be recycled or reused.

Nonprofit deconstruction companies keep the profits from the resale of salvaged materials and give the homeowner a tax write-off equal to the value of those materials. In this way, everyone benefits.

Deconstructing Deconstruction

If you're considering whether deconstruction is for you, cost, of course, is a major consideration. According to Primdahl, there are five significant factors that affect the costs of deconstruction as compared to demolition:

1. Site logistics. This includes the location and accessibility of the site and the layout of the house. For example, basements add a little bit more to the cost because they require another day and a half or so of labor.
2. Individual components of the building. Stucco, for instance, is much more difficult to deconstruct than lateral siding. Old houses are easier to deconstruct than new ones because they contain fewer adhesives and particle board, and because new nails are more difficult to extract.
3. Local disposal fees or tipping fees.
4. Construction crew experience.
5. The resale value of the salvageable material or the tax write-offs available when those materials are donated to a nonprofit building materials resale outlet.

Contrary to conventional expectations, deconstruction often turns out to be more affordable than traditional demolition. Two projects that were handled by the RE Stores in Washington State show two different ways that deconstruction can be profitable (see Table 1). In the first case, the bid cost was particularly low because the demolition bid was made by a relative of the homeowner, who would provide free labor; the listed bid consisted solely of projected disposal fee cost. Located in Anacortes, Washington, this house sat below the level of its U-shaped driveway. The owner wanted the foundations left intact for future use. Materials salvaged included a metal roof laid on 2 x 6 tongue-and-groove cardecking, exposed beams, a large cabinet set, nice siding, and wood paneling on most walls. The homeowner recently informed the company that, after taxes, he had saved $3,000 over what he would have spent on traditional machine demolition.

The second case involved a house that was located in Seattle, Washington, and had a narrow driveway that the owner wanted left intact. The house was made of brick and had a terra cotta tile roof. The owner saved almost $1,300 by choosing deconstruction rather than demolition and received a tax write-off of $10,000 for donating the salvaged materials.

Individual house savings can really add up when entire developments get a makeover. A private pilot project, which is being headed up by Primdahl, is leading to the deconstruction of a large housing development in Seattle. The development had 320 essentially identical units. Two of these were torn down; one was demolished and one was deconstructed. A precise comparison of the costs of the two methods was undertaken. The deconstruction cost $635 less than the demolition, in part because the owner diverted 30 tons of material from landfills.

Table 1. Breakdown of Deconstruction Costs

Case	Land-filled	Recycled	Reused	Disposal Cost	Labor	Total Cost	Tax Write-off	Demo Bid
1	11%	36%	53%	$1,600	$5,000 (250 hours)	$ 6,600	$19,000	$ 3,200-$ 3,800
2	20%	40%	40%	$4,150	$8,580 (390 hours)	$12,730	$10,000	$14,000

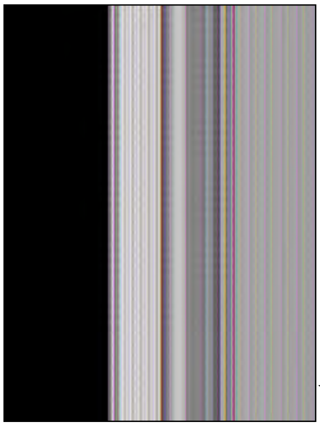
Many factors other than cost, however, can affect the decision to deconstruct. If components of an old house, such as antique wood paneling, are earmarked for reuse in a new house, deconstruction can do what demolition cannot. Deconstruction is also less likely to damage the surrounding greenery and can even avoid destroying the lawn. And in rainy areas such as Seattle, where demolition during rainfall can pollute the water supply, regulations are going into effect to prevent demolition on rainy days. As deconstruction during rainfall does not noticeably affect the water supply, these laws do not apply to deconstruction, and work on the site can continue unhindered by rain.

On the other hand, demolition has the advantage of speed. A demolition crew of four can take down a house in a day and a half; the same house would take a deconstruction crew of six about eight working days. In another case study provided by the Washington-based RE Stores, the deconstruction company's bid was equal to the demolition company's bid. The owner chose to go with the demolition company, because he was in a rush. Obviously, planning ahead to allow time for deconstruction can solve that problem, but even the best-laid plans can go awry.

Deconstruction for the Discerning Professional

"One thing that's missed by people who are considering getting into deconstruction is that the way it works best is if you have a retail outlet to sell the stuff to after you get it down," says Carl Weimer, executive director of Washington's RE Stores. As a professional who wants to take the plunge, you might be wondering what else you're missing.

To encourage hesitant professionals to rethink how they handle construction debris, a wide variety of programs have been springing into existence. Some aim to educate people in the industry about green building or construction waste management. Others offer incentives to encourage companies to get involved.



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A deconstruction worker carefully pries out a window.

The King County, Washington Construction Works program does both. When a company contacts the program regarding a job site, a Construction Works representative meets with the site superintendent and project engineer. The representative will help them write a construction waste management plan, calculate the project's recycling rate, and either set up an on-site recycling program or learn how to recycle more of their materials. They also receive copies of two guides: "Construction Recycling Directory" and "Contractors Guide: Save Money and Resources Through Job Site Recycling and Waste Prevention." If the company succeeds in recycling or reusing at least 60% of its waste at that job site, the county awards it a year's membership in Construction Works. The company receives a good deal of free publicity as a result. Until 2001, the requirements were 50% for commercial and 40% for residential job sites, but as technology has improved and environmental concern has deepened, the county has adopted the more stringent 60% requirement. In 2001 Construction Works had three members. That number jumped to six members in 2002, and by press time, the program had ten members, although the year is not yet half done.

The Green Building program, based in Alameda County, California, is tackling construction waste as part of a larger effort to minimize the impact of construction on the environment. Like other programs of its kind, Green Building offers training, publishes educational booklets, and holds workshops. The program's reach goes beyond education, extending into legislation and certification. Program managers work with local governments to develop and implement regulations to reduce the amount of waste going to landfills. And earlier this year, they formed a partnership with the San Francisco Bay Area chapter of the National Association of the Remodeling Industry to develop a green building certification program (see "Saving Resources with Green Construction," *HE* May/June '03, p. 12).

Iris Amdur of Natural Logic, Incorporated, is part of a construction waste management initiative in the Washington, D.C., area that straddles many barriers and groups. Its working name is Washington Construction Recycling Group, although that is not yet set in stone. This ambitious far-reaching plan aims not only to provide local resources for local contractors, but also to design a package that can be used to start similar programs all over the country. The group also plans to generate educational resources for people outside the industry, such as government officials and policymakers.

These are just three examples of the kinds of programs that are coming online to promote deconstruction and related techniques. These programs reflect a growing national awareness that efficient waste management is a necessity for professionals. Together, they are planning for a future when demolition, rather than deconstruction, will be the oddity.

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For more information:

Recycled Building Material Stores

Habitat for Humanity's Sound Builder's ReSource and the Building Materials Drop-Off Site in Olympia, Washington, www.sbroly.org

Habitat for Humanity Canada's ReStores, www.habitat.ca/restores.html

Recycle North in Vermont, www.recyclenorth.org/decon.html

RE Sources for Sustainable Communities and RE Stores in Washington, www.re-sources.org/restore/

The ReBuilding Center in Portland, Oregon, www.rebuildingcenter.org

Related Institutions

Alameda County Waste Management Authority's Green Building program, www.stopwaste.org

Associated General Contractors, www.agc.org/Environmental_Info/Recycle_This1.asp

Construction Materials Recycling Association, www.cdrecycling.org/wholeframe.htm

Deconstruction Institute, www.deconstructioninstitute.com

Institute for Local Self-Reliance, www.ilsr.org

King County Construction Works, www.metrokc.gov/greenworks

Peaks to Prairies Pollution Prevention Information Center, www.peakstoprairies.org

Portland Office of Sustainable Development, www.sustainableportland.org

Powell Center for Construction and Environment, www.cce.ufl.edu

U.S. Environmental Protection Agency Construction and Demolition Debris, www.epa.gov/epaoswer/non-hw/debris

Used Building Materials Association, http://bcn.boulder.co.us/environment/ubma/index.html

Used Building Materials Exchange, http://bcn.boulder.co.us/environment/ubma/ubma_tempexchange.htm

Washington Construction Recycling Group, contact Iris Amdur at (202)675-6940 or iris@natlogic.com

NARI (National Association of the Remodeling Industry), www.nari.org